Grade 10, Term 2, Lesson 1, Criterion C1
Outcomes, Support and Probability Mass Functions

Evaluated criteria — C1: Characterises discrete random variables in economic and financial contexts by identifying their support, probability mass function and possible outcomes.

Attempt–check routine. Attempt each task before consulting its worked solution. Work in your own notes before following a check or solution link. If you need help, consult just one stage, return to the task, and continue independently. Task links open the matching solution stage; return arrows lead back to the problem, its group, and the contents.

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1 Introduction

Problems:

1.1. Given outcomes and their probabilities

1.2. Outcomes from independent events

All data in this activity are hypothetical teaching models. Keep exact expressions where useful and round final probabilities to four decimal places.

Highlighted item	What it means	Usual symbols
Outcome	A possible result of the experiment.	ω (one outcome), Ω (sample space)
Probability	Probability expressions and PMF masses, without units.	$P(\omega)$, $p_X(x)$, q_i
First variable	The first variable, its numerical values, and any units.	X, x; F (fee)
Second variable	The second variable, its numerical values, and any units.	Y, y; E (electronic-payment indicator)
Third variable	The third variable, its numerical values, and any units.	Z, z; C (card-payment indicator)
Unknown parameter	A quantity to determine from the information given.	k

For variables, uppercase letters denote random variables; lowercase letters denote their possible values.

1.1 Given outcomes and their probabilities

A school shop receives one payment. A *random variable* assigns a number to each complete outcome. The table gives the outcome probabilities and maps each method to the fee F in USD, an electronic-payment indicator E, and a card-payment indicator C. Each indicator is 1 when its condition is met and 0 otherwise.

Task 1. Identify the outcomes and their probabilities. Read the four payment methods and their probabilities from the first two columns. The method observed on this payment is the complete outcome.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

Task 2. Define variables and units; map, list support, build and check PMFs. Use the supplied mappings; attempt these questions before following the check link.

1. Identify each variable's distinct supported values. State the units, or say unitless.
2. Which outcomes contribute to $P(F = 1)$? Explain why their probabilities are added.
3. Must E and C have identical PMFs because both have support $\{0, 1\}$? Use the mobile-wallet outcome to justify your decision.

Task 3. Find and interpret a requested probability. Calculate $P(F = 1)$ and interpret it for the shop.

Record your reasoning in your own notes.

Check after attempting.

Check: school-shop supports and PMFs

The *support* is the set of values with positive probability. Here it is $\{0, 1, 2\}$ for the fee in USD and $\{0, 1\}$ and $\{0, 1\}$ for the unitless indicators E and C, respectively.

A *probability mass function* (PMF) gives the probability of each value. Read the complete-outcome probabilities from the table and add those of the mutually exclusive outcomes that map to the same value. In particular, a 1 USD fee and a card payment each group debit and credit together. The electronic-payment group also includes the mobile wallet.

$$P(F = 1) = 0.2 + 0.3 = 0.5,$$
$$P(E = 1) = 0.2 + 0.3 + 0.4 = 0.9,$$
$$P(C = 1) = 0.2 + 0.3 = 0.5.$$

The fee PMF has masses $0.1, 0.5, 0.4$ at $0, 1, 2$. The indicators share a support but have different PMFs: mobile wallet maps to $E = 1$ and $C = 0$.

Indicator value (unitless)	0	1
$p_E(e)$	0.1	0.9
$p_C(c)$	0.5	0.5

Each PMF has non-negative masses totaling 1. Fee values have units of USD; all probabilities are unitless. See the worked grouping problem for the full fee calculation.

Task 3. Find and interpret a requested probability. Find and interpret the probability that the fee is 1 USD. From the fee PMF, $P(F = 1) = 0.5$. Half of payments incur a 1 USD processing fee in this model.



1.2 Outcomes from independent events

A school enterprise offers a 5 USD notebook to each of three visitors. For each visitor, the possible decisions are to buy (B, probability 0.4) or to pass (N, probability 0.6). Assume the visitors decide independently: one visitor's decision does not change the probabilities for the others.

Task 1. Identify the outcomes and their probabilities. Before looking at the eight outcomes, enumerate the ordered outcomes by buyer count. Explain what each position represents.

Buyers	Ordered outcomes (B or N in each position)
0	
1	
2	
3	

For BNN, assign one probability factor to each visitor's decision. What assumption justifies multiplying them?

$$P(\text{BNN}) = \underbrace{\quad}_{\text{visitor 1}} \times \underbrace{\quad}_{\text{visitor 2}} \times \underbrace{\quad}_{\text{visitor 3}}.$$

Complete the notebook mappings

X counts buyers (buyers); $Y = 5X$ is revenue (USD); Z is 1 if the first visitor buys and 0 otherwise (unitless).

Task 2. Define variables and units; map, list support, build and check PMFs. Complete one row per complete outcome. BNN is an illustrative row; check your enumeration against the first column.

Complete outcome	Outcome probability	X (buyers)	Y (USD)	Z (unitless)
NNN	_____			
BNN	0.144	1	5	1
NBN				
NNB				
BBN				
BNB				
NBB				
BBB				

Compare BNN and NBN: which assigned values agree, which differ, and why?

Now list each support without repeated values; keep units separate.

$S_X =$ _____ $S_Y =$ _____ $S_Z =$ _____

Task 3. Find and interpret a requested probability. Select *all* outcomes with exactly one buyer. Why does mutual exclusivity justify addition? Calculate and interpret $P(X = 1)$.

Diagnostic. A student writes $P(X = 1) = 0.4 \times 0.6 \times 0.6$. What event has the student actually calculated? Is it the whole event $X = 1$?

Build the remaining PMF entries and check their totals in your notebook. Then consult the linked check; complete mappings also appear there.

Check: notebook outcomes, mappings and grouping

The eight complete outcomes are

$$\Omega = \{\text{NNN}, \text{BNN}, \text{NBN}, \text{NNB}, \text{BBN}, \text{BNB}, \text{NBB}, \text{BBB}\}.$$

Multiply the three component probabilities to find the probability of one complete outcome. For example, in BNN only the first visitor buys, so

$$P(\text{BNN}) = 0.4 \times 0.6 \times 0.6 = 0.144.$$

The product has the same factors for NBN and NNB. The products are $0.6^3 = 0.216$ for NNN, $0.4^2 \times 0.6 = 0.096$ for each of BBN, BNB, and NBB, and $0.4^3 = 0.064$ for BBB.

The diagnostic product describes BNN: only the first visitor buys. It is one of three alternatives in $X = 1$, not their combined probability.

Complete outcome	Outcome probability	X (buyers)	Y (USD)	Z (unitless)
NNN	0.216	0	0	0
BNN	0.144	1	5	1
NBN	0.144	1	5	0
NNB	0.144	1	5	0
BBN	0.096	2	10	1
BNB	0.096	2	10	1
NBB	0.096	2	10	0
BBB	0.064	3	15	1

BNN and NBN both give one buyer and 5 USD. Their indicators differ because only BNN starts with a purchase. Reading distinct column values gives supports $\{0, 1, 2, 3\}$ (buyers), $\{0, 5, 10, 15\}$ (USD), and $\{0, 1\}$ (unitless).

To get the PMF entry for exactly one buyer, add the probabilities of the three mutually exclusive complete outcomes that give $X = 1$:

$$\begin{aligned} P(X = 1) &= P(\text{BNN}) + P(\text{NBN}) + P(\text{NNB}) \\ &= 0.144 + 0.144 + 0.144 = 0.432. \end{aligned}$$

In the order of their supports, X and Y both have masses $0.216, 0.432, 0.288, 0.064$. The indicator Z groups outcomes by their first letter, giving 0.600 at 0 and 0.400 at 1. The worked school-sale problem shows the complete mappings and sums.

The braces list the values in a support. A discrete support may be finite, as for the buyer count, or countably infinite, as in $\{1, 2, 3, \ldots\}$. **Classify and explain before reading on:** compare exact purchase time with purchase time rounded or recorded to whole seconds. Does using seconds determine the classification?

Check. Exact purchase time is continuous in an ideal measurement model: it can take values throughout an interval. Recording time in whole seconds gives discrete values. Classification concerns the modelled or recorded quantity, not merely its unit. No distribution for exact purchase time is supplied.

The notation reference shows how to write and check the sums used to build a PMF.

Task 3. Find and interpret a requested probability. Find and interpret the probability of exactly one buyer. Read $p_X(1) = 0.432$ from the buyer-count PMF. Exactly one of the three visitors buys in 43.2% of such sales.



2 Notation Reference and Practice Procedure

Problems:

2.1. Notation reference
2.2. Practice procedure

2.1 Notation reference

Symbol	Meaning and units	Example from the school shop	Example from the school sale
ω	One complete experimental outcome	Debit card is one complete outcome.	In BNN, only the first visitor buys.
Ω	Sample space, the set of all complete outcomes	{Cash, debit card, credit card, mobile wallet}	{NNN, BNN, NBN, NNB, BBN, BNB, NBB, BBB}
X, x	The rule that assigns a number and a particular numerical value, respectively	F gives the fee in USD, and 1 is one possible fee value.	X counts buyers, and $x = 1$ means one buyer.
$X(\omega)$	The numerical value assigned to outcome ω	$F(\text{debit card}) = 1$	$X(\text{BNN}) = 1$
S	Support, the set of values with positive probability	$S = \{0, 1, 2\}$ for the fee in USD.	$S = \{0, 1, 2, 3\}$ for the buyer count.
$p_X(x)$	Probability assigned to value x, with no units	$p_F(1) = P(F = 1) = 0.5$, the probability of a 1 USD fee.	$p_X(1) = P(X = 1)$, the chance of exactly one buyer.
$\sum$	Add the indicated terms	Add the probabilities of debit card and credit card. $\sum_{\omega : F(\omega)=1} P(\omega) = 0.2 + 0.3 = 0.5$	$\sum_{X(\omega)=1} P(\omega)$ adds the probabilities of BNN, NBN, and NNB.

Read an outcome and its value. The lowercase Greek letter ω (omega) stands for one complete outcome. The capital Ω names the whole sample space. Read $\omega \in \Omega$ as *omega belongs to Omega*. Read $X(\omega)$ as *X of omega*, the value the rule X assigns to that outcome.

The *probability mass function* (PMF) assigns a probability to each value of a discrete random variable. For X, it is written $p_X(x) = P(X = x)$, read "the probability that X equals x". The subscript identifies the random variable: $p_Y(y) = P(Y = y)$ describes revenue, while $p_Z(z) = P(Z = z)$ describes the first visitor's purchase indicator. The expression $X = x$ describes the event that the variable takes the chosen value.

Find one PMF entry. Keep one row per complete outcome in a mapping table, then one entry per distinct supported value in the PMF. Fix a numerical value x and keep every outcome that maps to it.

$$p_X(x) = \sum_{\omega \in \Omega : X(\omega) = x} P(\omega).$$

Read the colon as *such that* and the capital sigma $\sum$ as *add*. The condition under the sum says which outcomes to include. Each $P(\omega)$ is the probability of one complete outcome. Those outcomes are mutually exclusive, so add their probabilities once each, even when the probabilities differ. An empty sum is zero.

Verify the PMF. Each mass must satisfy $p_X(x) \geq 0$, where $\geq$ means *greater than or equal to*. The support S contains precisely the values with positive mass. Every value outside S has mass zero. To check *normalization*, add the masses once per supported numerical value.

$$\sum_{x \in S} p_X(x) = 1.$$

Here $x \in S$ means x *belongs to* S. The events for distinct values cannot occur together, and the supported values account for all probability. The total of 1 represents a 100% chance of obtaining some supported value.

Finite and countably infinite supports. A finite sum has a last term. For a countably infinite support such as $S = \{1, 2, 3, \ldots\}$, the same notation represents an infinite list of non-negative terms.

$$\sum_{x \in S} p_X(x) = p_X(1) + p_X(2) + p_X(3) + \cdots = 1.$$

Check that the partial sums, the totals of the first finitely many masses, approach 1 as more terms are included. An infinite list of possible values can have total probability 1.

See the grouping example for both sums in the same model.



2.2 Practice procedure

Reasoning chain: complete outcomes and probabilities → variable definition and units → mappings → support → PMF → validity checks → requested probability and contextual interpretation.

Task 1. Identify the outcomes and their probabilities. List all complete outcomes. Read supplied probabilities, or list ordered decisions and multiply the component probabilities within each independent outcome. For a supplied PMF, identify the observed categories or counts. Recover any missing outcome probability by making the total 1.

Task 2. Define variables and units; map, list support, build and check PMFs. Define each discrete variable and its units. Map each outcome to a value and list the distinct values with positive mass as the support. Keep one row per complete outcome in your mapping. To build the PMF, use one entry per distinct supported value and add the probabilities of mutually exclusive outcomes that give that value. Check that the masses are non-negative and total 1.

Task 3. Find and interpret a requested probability. Write the event in terms of the specified variable, paying attention to endpoints. Use the PMF to find its probability, keep intermediate values exact, and interpret the result in context.



3 General Worked Problems — Given Outcomes or PMFs

Problems:

3.1. Introductory general problem: grouping outcomes

3.2. Payment methods and a fee

3.3. Daily insurance claims

3.4. Investment results with a missing mass

When complete-outcome probabilities are given, add the masses of the mutually exclusive outcomes that map to each value. When a PMF is supplied with one missing mass, use the total of 1 to recover it.

3.1 Introductory general problem: grouping outcomes

This complete worked model develops the fee model from the first Introduction example.

Mathematical wording

An experiment has outcomes a, b, c, d with probabilities 0.1, 0.2, 0.3, 0.4. Define $X(a) = 0$, $X(b) = X(c) = 1$, $X(d) = 2$.

Questions/Tasks



C1 Task 1. Identify the outcomes and their probabilities. Record the four complete outcomes and the probability of each.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define the score X, map each outcome to its score, list the support, and build and check the PMF.

C1 Task 3. Find and interpret a requested probability. Use the PMF to find $P(X = 1)$ and explain what it means for the score.

Choose the mathematical wording or one financial version below; these are alternative wordings of the same model. Each task links to the matching part of the shared solution.

Description	Contextualized task
Version 1 — School shop processing fees. A school shop receives one payment. There are exactly four possible payment methods: cash, debit card, credit card, or a mobile wallet. The chance of cash is 10%, of debit card is 20%, of credit card is 30%, and of a mobile wallet is 40%. The shop pays no processing fee for cash, a fee of $1 for either type of card, and a fee of $2 for a mobile wallet. These are the only processing charges.	**Task 1. Identify the outcomes and their probabilities.** Record each payment method and its probability. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the processing fee with units. Map the methods to their fees, list the support, and group the probabilities to build and check the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of a $1 processing fee. Interpret the result.
Version 2 — Bike rental late fees. A bike rental shop charges a late fee based on when a bike is returned. A rental ends on time or early with probability 10%, 1 to 10 minutes late with probability 20%, 11 to 20 minutes late with probability 30%, or more than 20 minutes late with probability 40%. The shop records lateness in whole minutes. The fee is $0 for an on-time or early return, $1 for either of the two shorter delays, and $2 for a delay of more than 20 minutes.	**Task 1. Identify the outcomes and their probabilities.** Record each return category and its probability. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the late fee with units. Map the return categories to their fees, list the support, and group the probabilities to build and check the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of a $1 late fee. Interpret the result.
Version 3 — Recycling buyback payments. At a school recycling drive, each delivery contains one type of material. The probabilities are 10% for non-recyclable material, 20% for paper, 30% for plastic, and 40% for aluminium. These are the only delivery types. A recycling centre pays the school $0 for a non-recyclable delivery, $1 for a delivery of either paper or plastic, and $2 for an aluminium delivery.	**Task 1. Identify the outcomes and their probabilities.** Record each delivery type and its probability. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the payment to the school with units. Map each type to its payment, list the support, and group the probabilities to build and check the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance the school receives $1. Interpret the result.

Complete worked solution

Task 1. Identify the outcomes and their probabilities.

The complete outcomes a, b, c, d are mutually exclusive and exhaustive. The table gives their probabilities and names the corresponding outcomes in each version.

	a	b	c	d
Probability	0.1	0.2	0.3	0.4
Version 1	Cash	Debit card	Credit card	Mobile wallet
Version 2	On time or early	1–10 min late	11–20 min late	Over 20 min late
Version 3	Non-recyclable	Paper	Plastic	Aluminium

Task 2. Define variables and units; map, list support, build and check PMFs.

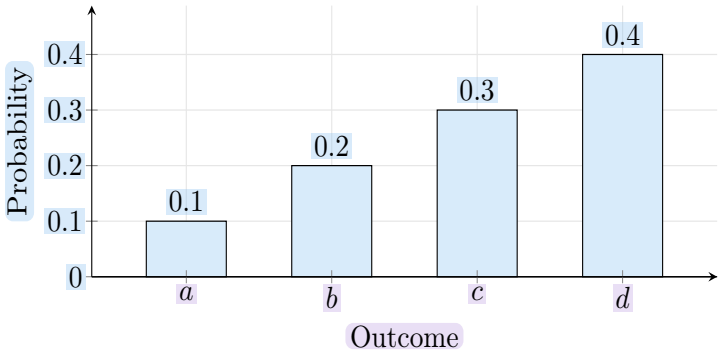
X is a score in the abstract model. It measures the shop's processing fee in Version 1, the renter's late fee in Version 2, and the school's payment per delivery in Version 3, all in USD. In each case the mapping is

$$X(a) = 0, \qquad X(b) = X(c) = 1, \qquad X(d) = 2.$$

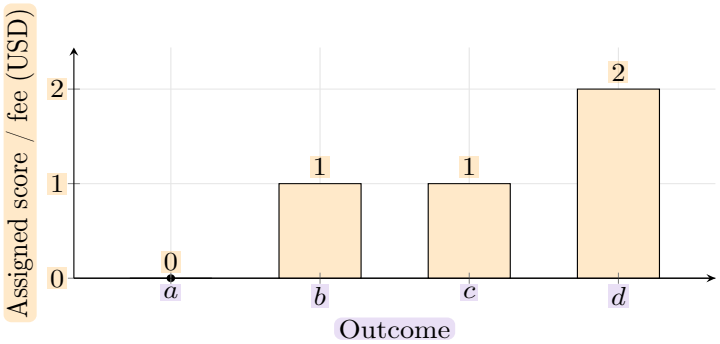
The support is $\{0, 1, 2\}$. Only a gives 0 and only d gives 2. Both b and c give 1, so add their probabilities once each.

x	0	1	2
$p_X(x)$	0.1	$0.2 + 0.3 = 0.5$	0.4

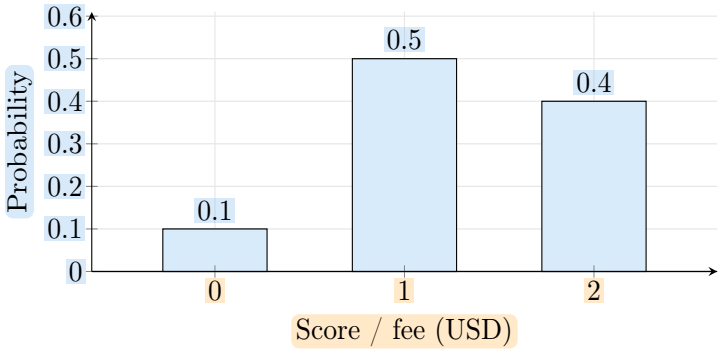
All masses are non-negative and sum to $0.1 + 0.5 + 0.4 = 1$. A value outside the support, such as 3, has mass zero because no outcome maps to it.



Outcome probabilities. Distinct outcomes have their own chances.



Assigned values (Version 1). Both b and c give 1; in the financial wording these are fees in USD.



Grouped PMF (Version 1 fee scale). The probability at 1 is 0.2 + 0.3 = 0.5.

Task 3. Find and interpret a requested probability.

Read the mass at 1 from the PMF: $P(X = 1) = p_X(1) = 0.5000$. The score equals 1 in half of repetitions. The probability is 50% that the shop pays a \$1 processing fee, the renter pays a \$1 late fee, or the school receives \$1, in Versions 1, 2, and 3 respectively.

3.2 Payment methods and a fee

Partially guided attempt. Use the given probabilities with a different fee schedule. Complete the tables and Tasks 1–3 before following the task links to the worked solution.

A shop receives one payment: cash with probability 0.25, debit with 0.45, or credit with 0.30. Cash costs no processing fee; either card costs 1 USD.

Questions/Tasks

C1 Task 1. Identify the outcomes and their probabilities. Record the complete payment-method outcomes and their probabilities.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define the fee variable with units, map each payment method to its fee, and list the support. Build and check the PMF. Explain why three outcomes give only two fee values.

C1 Task 3. Find and interpret a requested probability. Use the PMF to find the probability of a positive processing fee and interpret it for the shop.

Complete outcome	Probability	Fee X (USD)
Cash	0.25	0
Debit	_____	_____
Credit		

Fee (USD)	Matching outcomes	Sum of probabilities
_____	_____	_____

In your notes, define X, give its support, check the PMF, and interpret your requested probability.

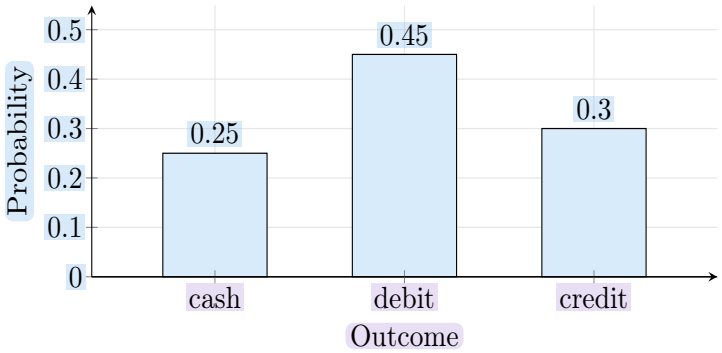
Complete worked solution

Task 1. Identify the outcomes and their probabilities.

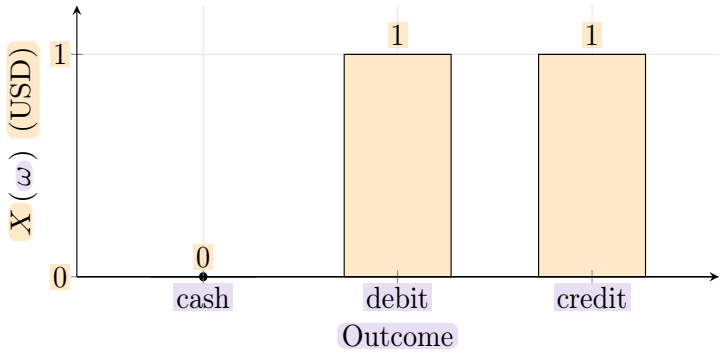
The complete outcomes are cash, debit, and credit, with probabilities 0.25, 0.45, 0.30, respectively.

Task 2. Define variables and units; map, list support, build and check PMFs.

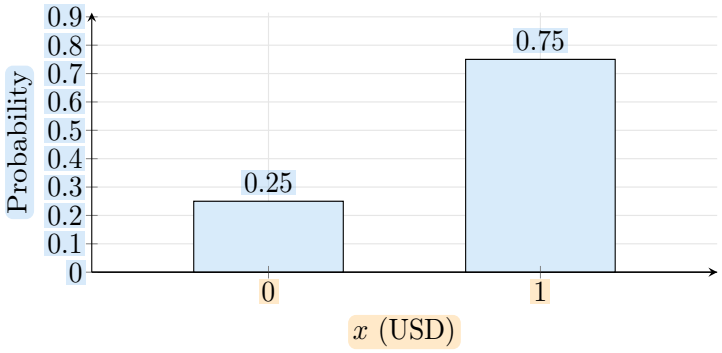
X measures the processing fee in USD. Cash gives 0 and each card method gives 1, so the support is $\{0, 1\}$. Only cash contributes to $p_X(0) = 0.25$. Both cards give the same value, so $p_X(1) = 0.45 + 0.30 = 0.75$. Both masses are non-negative and sum to 1.



Payment-method probabilities. The bars refer to method outcomes.



Assigned fees. Both cards map to 1 USD; cash maps to zero.



Grouped fee PMF. Debit and credit contribute $0.45 + 0.30 = 0.75$ at 1 USD.

Task 3. Find and interpret a requested probability.

The only positive fee is 1 USD, so $P(X > 0) = p_X(1) = 0.7500$. The shop incurs a processing fee on 75% of payments.

3.3 Daily insurance claims

The supplied PMF has one missing mass. Use normalization to recover it.

Observe a small policy group for one day. The complete outcome is the observed claim count: no claims, one claim, or two claims, with probabilities 0.7, q, 0.1, respectively.

Questions/Tasks

C1 Task 1. Identify the outcomes and their probabilities. List the possible observed claim counts and their probabilities. Use their total to find q.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define X with units and map each observed count to its value. List the support and build and verify the completed PMF.

C1 Task 3. Find and interpret a requested probability. Use the PMF to find the probability of at least one claim and interpret it for the policy group.

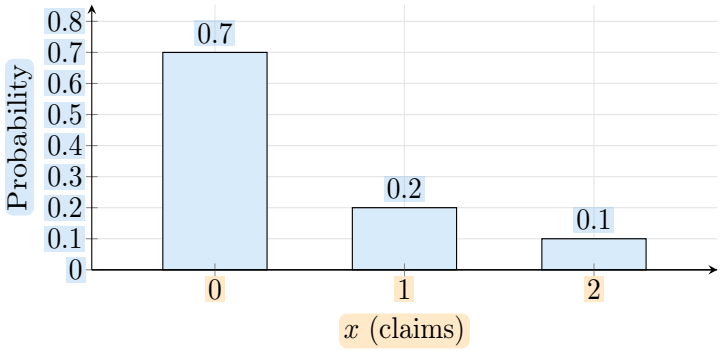
Complete worked solution

Task 1. Identify the outcomes and their probabilities.

No claims has probability 0.7, and two claims has probability 0.1. These outcomes and one claim exhaust the possibilities, so one claim has probability $q = 1 - 0.7 - 0.1 = 0.2$.

Task 2. Define variables and units; map, list support, build and check PMFs.

X counts claims in one day, in claims. No claims, one claim, and two claims map to $0, 1, 2$, respectively. Each value comes from one outcome. On support $\{0, 1, 2\}$, the masses are 0.7, 0.2, 0.1 in order. All are positive and total 1.



Completed claim-count PMF. The solved parameter is $q = 0.2$; its known mass is 0.2.

Task 3. Find and interpret a requested probability.

$P(X \geq 1) = p_X(1) + p_X(2) = 0.2 + 0.1 = 0.3000$. The group has at least one claim on 30% of days in this model.

3.4 Investment results with a missing mass

The supplied PMF has one missing mass. Use normalization to test the proposed value and recover the correct mass.

Observe the net result of a hypothetical investment at the end of one month. The complete outcome is the observed gain category: a 20 USD loss, break-even, or a 30 USD gain, with probabilities 0.2, 0.5, q, respectively. A student proposes $q = 0.4$.

Questions/Tasks

C1 Task 1. Identify the outcomes and their probabilities. Record the possible gain categories and their probabilities. Test the proposed q and recover the missing probability by normalization.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define the net gain G with units. Map each category to its signed value, list the support, and build and check the corrected PMF.

C1 Task 3. Find and interpret a requested probability. Use the PMF to find the probability of a non-negative gain and interpret it for the investment.

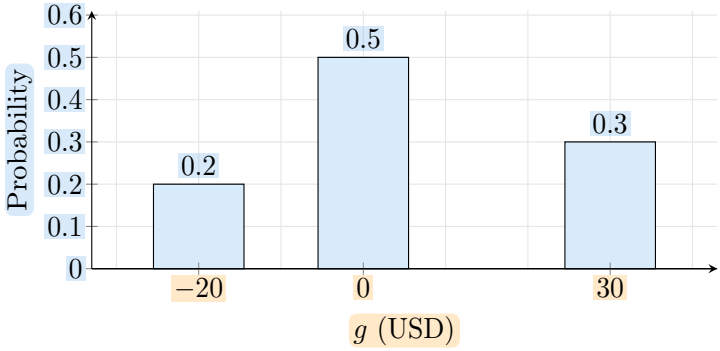
Complete worked solution

Task 1. Identify the outcomes and their probabilities.

The loss and break-even outcomes have probabilities 0.2 and 0.5. The proposed 0.4 for the gain would make the total 1.1. Instead, the gain has probability $q = 1 - 0.2 - 0.5 = 0.3$.

Task 2. Define variables and units; map, list support, build and check PMFs.

G is the one-month net gain in USD. Loss, break-even, and gain map to $-20, 0, 30$, respectively. These three amounts form the support. Each has one outcome, giving masses 0.2, 0.5, 0.3, respectively. The corrected masses are non-negative and sum to 1.



Corrected gain PMF. The solved parameter is $q = 0.3$; the bar at 30 USD uses known mass 0.3. Gain values retain their numerical spacing.

Task 3. Find and interpret a requested probability.

Include break-even in a non-negative gain:

$$P(G \geq 0) = p_G(0) + p_G(30) = 0.5 + 0.3 = 0.8000.$$

The investment breaks even or makes a gain with probability 80%.



4 General Worked Problems — Independent Events

Problems:

4.1. School sale: three probability mass functions

4.2. Notebook buyers

Multiply the probabilities of independent decisions within each complete outcome. Then add the probabilities of the mutually exclusive complete outcomes that map to each value.

4.1 School sale: three probability mass functions

Return to the second Introduction example, where we built the ordered outcomes for the three-visitor school sale. Use the buyer count X, the revenue Y, and the first-visitor indicator Z from the Introduction.

For each visitor in the school sale, assign

$$P(\mathrm{B}) = 0.4, \qquad P(\mathrm{N}) = 0.6,$$

where B means buying a notebook and N means not buying. These probabilities add to 1. Assume that the three visitors decide independently and that each has these same probabilities. Independence means that one visitor's decision does not change the probabilities for another visitor.

Questions/Tasks



C1 Task 1. Identify the outcomes and their probabilities. List all eight complete outcomes in visitor order and multiply within each outcome to calculate its probability.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define X, Y, and Z with their units where applicable. Map each ordered outcome to its three values, list the supports, and build and check all three PMFs.

C1 Task 3. Find and interpret a requested probability. Use the revenue PMF to find $P(Y = 5)$ and interpret it for the school sale.

Complete worked solution

Task 1. Identify the outcomes and their probabilities.

As in the Introduction, independence allows us to multiply the probabilities of the three decisions within each ordered outcome. For example,

$$P(\text{BNN}) = 0.4 \times 0.6 \times 0.6 = 0.144,$$
$$P(\text{BBN}) = 0.4 \times 0.4 \times 0.6 = 0.096.$$

Applying the same rule to all eight outcomes gives

$$P(\text{NNN}) = 0.6^3 = 0.216, \qquad P(\text{BNN}) = 0.4 \times 0.6^2 = 0.144,$$
$$P(\text{NBN}) = 0.6 \times 0.4 \times 0.6 = 0.144, \qquad P(\text{NNB}) = 0.6^2 \times 0.4 = 0.144,$$
$$P(\text{BBN}) = 0.4^2 \times 0.6 = 0.096, \qquad P(\text{BNB}) = 0.4 \times 0.6 \times 0.4 = 0.096,$$
$$P(\text{NBB}) = 0.6 \times 0.4^2 = 0.096, \qquad P(\text{BBB}) = 0.4^3 = 0.064.$$

Several complete outcomes share a probability, and the probabilities vary with the number of buyers. Check the total by adding the probabilities of the eight mutually exclusive outcomes.

$$0.216 + 3(0.144) + 3(0.096) + 0.064 = 1.$$

Task 2. Define variables and units; map, list support, build and check PMFs.

X counts buyers, $Y = 5X$ measures revenue in USD, and Z is 1 if the first visitor buys and 0 otherwise, with no units. Their supports are $\{0,1,2,3\}$, $\{0,5,10,15\}$, and $\{0,1\}$, respectively. For each value, add the probabilities of the mutually exclusive complete outcomes that give it. Each product in Task 1 describes one sequence. Each sum here combines alternative sequences.

- **Number of buyers.** The outcomes BNN, NBN, and NNB produce $X = 1$, so

$$\begin{aligned} p_X(1) &= P(\text{BNN}) + P(\text{NBN}) + P(\text{NNB}) \\ &= 0.144 + 0.144 + 0.144 = 0.432. \end{aligned}$$

  Similarly, BBN, BNB, and NBB produce $X = 2$, so

$$p_X(2) = 0.096 + 0.096 + 0.096 = 0.288.$$

  Only NNN gives $X = 0$, and only BBB gives $X = 3$. The complete PMF is

$$p_X(0) = 0.216, \quad p_X(1) = 0.432,$$
$$p_X(2) = 0.288, \quad p_X(3) = 0.064.$$

All four count masses are non-negative and $0.216 + 0.432 + 0.288 + 0.064 = 1$.

Predict before checking the other PMFs. Use the count PMF to predict all revenue masses. Explain why $P(Y = 10) = P(X = 2)$ although their supports and units differ. Then group the outcomes with $Z = 1$: what must their sum be to agree with the first visitor's given purchase probability?

Check your revenue and indicator predictions.

Check: corresponding masses and the first visitor

- **Sales revenue.** Each notebook sells for 5 USD, so $Y = 5X$ when revenue is recorded in USD. The same outcomes that produce one buyer produce revenue of 5 USD. Thus the complete PMF is

$$p_Y(0) = 0.216, \quad p_Y(5) = 0.432,$$
$$p_Y(10) = 0.288, \quad p_Y(15) = 0.064.$$

  The revenue values have monetary units; their probabilities do not. In particular, $Y = 10$ and $X = 2$ select exactly BBN, BNB, and NBB, so both probabilities are 0.288.

- **First visitor's purchase.** The first visitor buys in BNN, BBN, BNB, and BBB, so

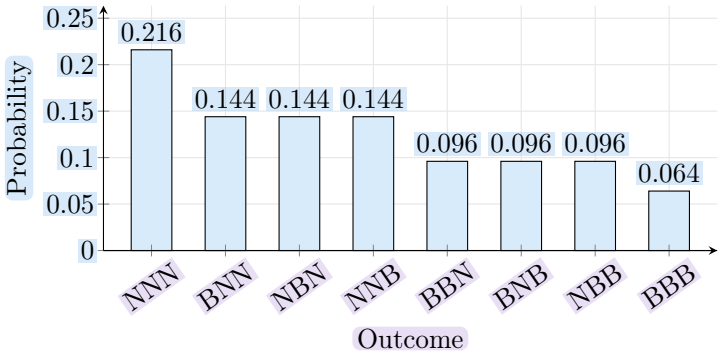
$$\begin{aligned} p_Z(1) &= 0.144 + 0.096 + 0.096 + 0.064 \\ &= 0.400. \end{aligned}$$

  The first visitor does not buy in NNN, NBN, NNB, and NBB, so

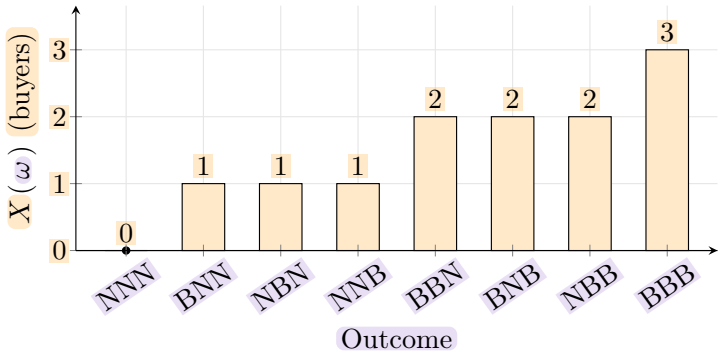
$$\begin{aligned} p_Z(0) &= 0.216 + 0.144 + 0.144 + 0.096 \\ &= 0.600. \end{aligned}$$

For X and Y, the probabilities total $0.216 + 0.432 + 0.288 + 0.064 = 1$; for Z, they total $0.600 + 0.400 = 1$. All entries are non-negative, and values outside each support have mass zero. The probabilities for Z agree with the original chances of the first visitor not buying or buying. Although X and Y have different supports and units, corresponding values have the same probabilities because $Y = 5X$. The variable Z groups the outcomes differently.

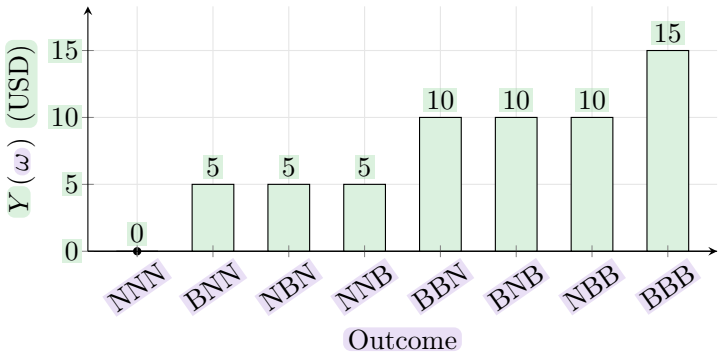
The events $X = 1$ and $Y = 5$ group the same three outcomes. The event $Z = 1$ groups the four outcomes that begin with B.



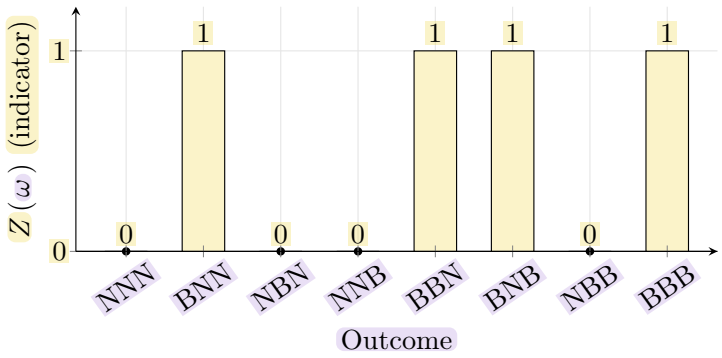
School sale: outcome probabilities. Each bar describes one complete three-visitor outcome.



Assigned buyer counts. The height is $X(\omega)$ in buyers, not a probability.



Assigned revenue. The height is $Y(\omega) = 5\,X(\omega)$ in USD.



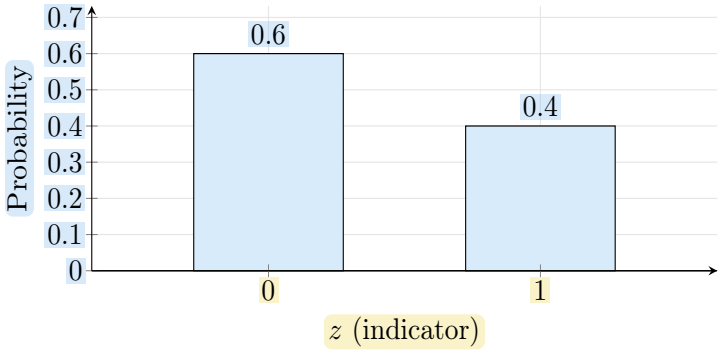
Assigned indicator. $Z(\omega)$ is 1 exactly when the first visitor buys; it has no units.



Buyer-count PMF. Group outcomes by X and add their probabilities.



Revenue PMF. Group outcomes by Y; probabilities have no USD units.



First-visitor PMF. Group outcomes by Z; these bars give probabilities of indicator values.

Task 3. Find and interpret a requested probability.

The revenue PMF gives $P(Y = 5) = p_Y(5) = 0.4320$. The school takes in exactly 5 USD with probability 43.2%.

4.2 Notebook buyers

Independent attempt. This time, there are two visitors and a different price per sale. Choose your own representation and complete the full reasoning chain in Tasks 1–3 before consulting any solution. Record your full working in your own notes.

Two visitors buy independently with probability $1/2$ each. Let B mean a purchase and N no purchase. Each sale earns 4 USD.

Questions/Tasks



C1 Task 1. Identify the outcomes and their probabilities. List all ordered purchase decisions and calculate the probability of each complete outcome.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define the buyer count X and the revenue Y, with units. Map each ordered outcome to both values, list the supports, and build and check both PMFs.

C1 Task 3. Find and interpret a requested probability. Use the revenue PMF to find the probability of earning exactly 4 USD. Interpret the result.

Before following a solution link, check that you included outcomes and probabilities, definitions and units, mappings, supports, PMFs, validity checks, and the requested probability with its interpretation.

Complete worked solution

Task 1. Identify the outcomes and their probabilities.

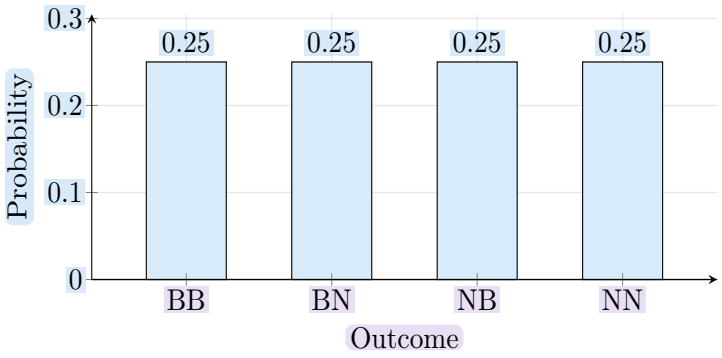
Both decisions have probability $1/2$ for each visitor. By independence, each ordered outcome BB, BN, NB, and NN has probability $(1/2)^2 = 1/4$.

Task 2. Define variables and units; map, list support, build and check PMFs.

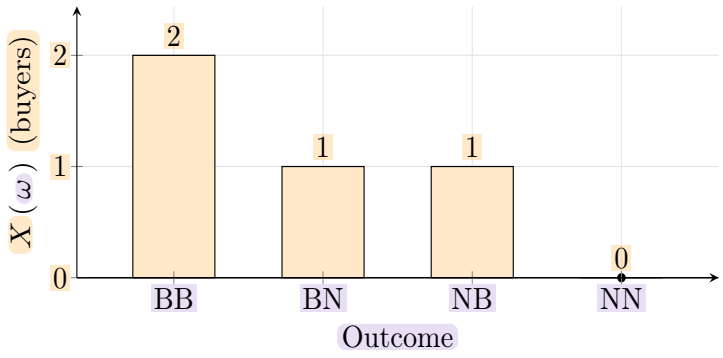
X counts buyers and $Y = 4X$ measures revenue in USD. For BB, BN, NB, and NN in order, the counts are $2, 1, 1, 0$ and the revenues are $8, 4, 4, 0$. The supports are $\{0, 1, 2\}$ and $\{0, 4, 8\}$. Both BN and NB give one buyer and 4 USD in revenue, so add $1/4$ and $1/4$ for the middle entry of each PMF. Each other value comes from one outcome.

x (buyers)	0	1	2
$p_X(x)$	0.25	0.50	0.25
y (USD)	0	4	8
$p_Y(y)$	0.25	0.50	0.25

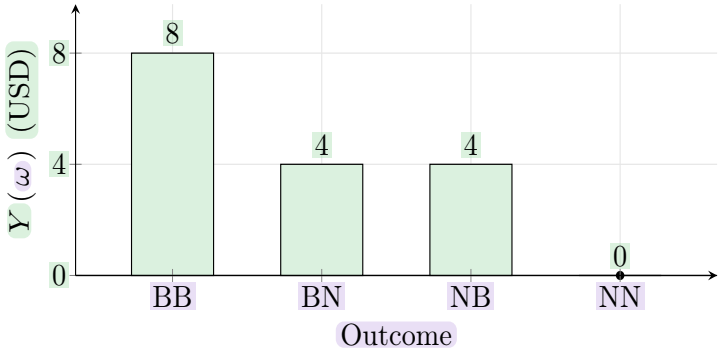
Both rows of masses are non-negative and total 1.



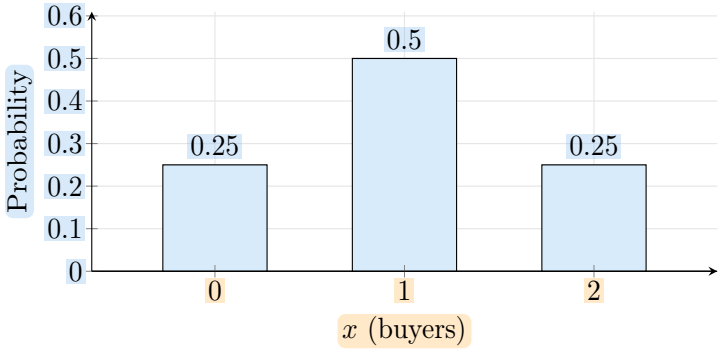
Complete-outcome probabilities. Each outcome has the same product of two decision probabilities.



Assigned buyer counts. The bars show $X(\omega)$, measured in buyers.



Assigned revenue. Each outcome maps to $Y(\omega) = 4\,X(\omega)$ in USD; these heights are assigned values.



Buyer-count PMF. BN and NB are grouped at one buyer.



Revenue PMF. $Y = 4\,X$ in USD; the corresponding probabilities are unchanged.

Task 3. Find and interpret a requested probability.

$P(Y = 4) = p_Y(4) = P(X = 1) = 0.5000$. The chance of taking in exactly 4 USD from the two visitors is 50%.

5 Advanced Problems

Problems:

5.1. Advanced general problem: a finite model

5.2. Advanced general problem: an infinite model

5.3. Calls until a sale

5.4. Delivery records: two numerical descriptions

5.1 Advanced general problem: a finite model

Mathematical wording

The masses in this model are given in proportion to one another. Use normalization to determine their common scale.

Run the experiment once and observe its score category: zero, two, or five, with probabilities c, $2c$, $3c$, respectively. Let X be the numerical score.

Questions/Tasks



C1 Task 1. Identify the outcomes and their probabilities. List the score categories. Use normalization to find c and the probability of each outcome.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Map the categories to scores, list the support, and build and verify the PMF of X.

C1 Task 3. Find and interpret a requested probability. Use the PMF to find $P(X \geq 2)$ and interpret the result for the score.

The table presents the same model in three financial settings. Each task links to the matching part of the shared solution.

Description	Contextualized task
Version 1 — Promotional refunds. A shop offers a promotional refund on one purchase. The refund can only be $0, $2, or $5. A $2 refund is twice as likely as no refund, and a $5 refund is three times as likely as no refund. These are the only possible refund amounts.	**Task 1. Identify the outcomes and their probabilities.** Record the refund categories. Use the relative likelihoods and normalization to find each category's probability. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the refund with units, map the categories to amounts, and list the support. Build and verify the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of a refund of at least $2. Interpret the result.
Version 2 — Fundraising draw prizes. At a school fundraiser, a participant draws one ticket from a box. A ticket pays $0, $2, or $5. The box contains twice as many $2 tickets as no-prize tickets and three times as many $5 tickets as no-prize tickets. Each ticket has the same chance of being drawn.	**Task 1. Identify the outcomes and their probabilities.** Record the observed ticket payout categories. Use the ticket ratios and normalization to find each category's probability. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the payout with units, map the categories to amounts, and list the support. Build and verify the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of winning at least $2. Interpret the result.
Version 3 — Equipment maintenance costs. A student printing business sets aside money for printer maintenance. One inspection can lead to no repair at a cost of $0, an adjustment costing $2, or a replacement part costing $5. Exactly one result occurs per inspection. An adjustment is twice as likely as no repair, and a replacement is three times as likely as no repair.	**Task 1. Identify the outcomes and their probabilities.** Record the inspection results. Use their relative likelihoods and normalization to find each outcome probability. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the maintenance cost with units, map the results to costs, and list the support. Build and verify the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of a maintenance cost of at least $2. Interpret the result.

Complete worked solution

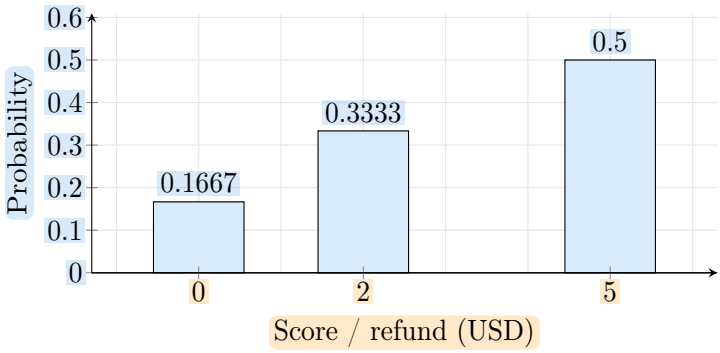
Task 1. Identify the outcomes and their probabilities.

The table lists the complete observed categories in each version. Within a row, the three probabilities are c, $2c$, $3c$ in order. Since these outcomes exhaust the possibilities, $c + 2c + 3c = 6c = 1$. Thus $c = 1/6$ and the three probabilities in order are $1/6$, $1/3$, $1/2$.

Model	First category	Second category	Third category
Score	Zero	Two	Five
Version 1	No refund	2 USD refund	5 USD refund
Version 2	No-prize ticket	2 USD ticket	5 USD ticket
Version 3	No repair	Adjustment	Replacement part

Task 2. Define variables and units; map, list support, build and check PMFs.

X measures the score in the abstract model. In Versions 1, 2, and 3 it measures the refund, prize payout, and maintenance cost, respectively, in USD. The three categories in each row map to $0, 2, 5$ in order. The support is $\{0, 2, 5\}$. There is one outcome per value, so the PMF has masses $1/6$, $1/3$, $1/2$ at those values. All are positive and sum to 1. The gap between 2 and 5 contains no supported values.



Normalized finite PMF (Version 1 refund scale). Only 0, 2, and 5 are supported; the gap is preserved. Heights display rounded probabilities.

Task 3. Find and interpret a requested probability.

$P(X \geq 2) = p_X(2) + p_X(5) = 1/3 + 1/2 = 5/6 \approx 0.8333$. The chance of a score of at least 2 is about 83.33%. In Versions 1, 2, and 3, this is the chance of a refund, payout, or maintenance cost of at least \$2, respectively.

5.2 Advanced general problem: an infinite model

Mathematical wording

Repeat independent trials, each with success probability $1/2$, and stop at the first success. Let S stand for success and F for failure. The supplied model assigns $p_T(t) = 2^{-t}$ for $t = 1, 2, \ldots$, where T is the trial count through the first success.

Questions/Tasks



C1 Task 1. Identify the outcomes and their probabilities. List the complete stopping sequences and use independence to give the probability of each.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define T, map the stopping sequences to trial counts, and list the infinite support. Build the PMF and verify the supplied model.

C1 Task 3. Find and interpret a requested probability. Use the PMF to find $P(T > 3)$ and interpret it in terms of trials.

The table presents the same model in three financial settings. Each task links to the matching part of the shared solution.

Description	Contextualized task
Version 1 — Calls until the first sale. A student running a school enterprise calls potential customers until the first sale. Each call is an independent attempt, with a constant one-half chance of making a sale. Count every call, including the call that makes the first sale. There is no limit on the number of calls allowed, and calling stops at the first sale.	**Task 1. Identify the outcomes and their probabilities.** Use S for a sale and F for a failed call. List the stopping sequences and calculate their probabilities. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the call count through the first sale. Map the sequences to counts, list the infinite support, and build and verify the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of needing more than 3 calls. Interpret the result.
Version 2 — Finding a quote within budget. A student business requests supplier quotes one at a time and stops at the first quote within its budget. Each quote has a one-half chance of meeting the budget, independent of the other quotes. There is no fixed limit on the number of quotes. Count each quote received, including the first one within budget.	**Task 1. Identify the outcomes and their probabilities.** Use S for a quote within budget and F for a quote above budget. List the stopping sequences and calculate their probabilities. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the quote count through the first one within budget. Map the sequences to counts, list the infinite support, and build and verify the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of needing more than 3 quotes. Interpret the result.
Version 3 — Finding an invoice with a discount. A bookkeeping trainee samples past supplier invoices to find an example of an early-payment discount. Exactly half of the invoices qualify for the discount. On each draw the trainee picks an invoice uniformly at random and puts it back after a miss. The draws are independent, and the chance of success on each draw stays one-half. There is no fixed limit on the number of draws. The trainee stops at the first qualifying invoice and includes that draw in the count.	**Task 1. Identify the outcomes and their probabilities.** Use S for a qualifying invoice and F for a miss. List the stopping sequences and calculate their probabilities. **Task 2. Define variables and units; map, list support, build and check PMFs.** Define the draw count through the first qualifying invoice. Map the sequences to counts, list the infinite support, and build and verify the PMF. **Task 3. Find and interpret a requested probability.** Use the PMF to find the chance of needing more than 3 draws. Interpret the result.

Complete worked solution

Task 1. Identify the outcomes and their probabilities.

The complete stopping outcomes are S, FS, FFS, ... The outcome ending on trial t is $t-1$ failures followed by one success. Independence gives its probability as $(1/2)^{t-1}(1/2) = 2^{-t}$. Thus the first three outcomes have probabilities $1/2, 1/4, 1/8$. Here S means a sale in Version 1, a quote within budget in Version 2, and a qualifying invoice in Version 3. F means a miss in each case. The probability rule applies to all three versions.

Task 2. Define variables and units; map, list support, build and check PMFs.

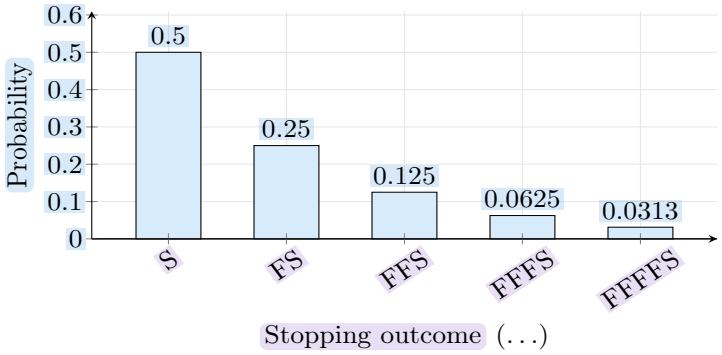
T counts attempts including the first success, in trials in the abstract model and in calls, quotes, or invoice draws in Versions 1, 2, and 3, respectively. The outcomes S, FS, FFS, and so on map to $1, 2, 3, \ldots$. There is one sequence of length t, so $p_T(t) = 2^{-t}$ on the countably infinite support $\{1, 2, \ldots\}$.

All masses are positive. To check their sum, use $2^{-t} = 2^{-(t-1)} - 2^{-t}$ and cancel consecutive terms through any finite count m:

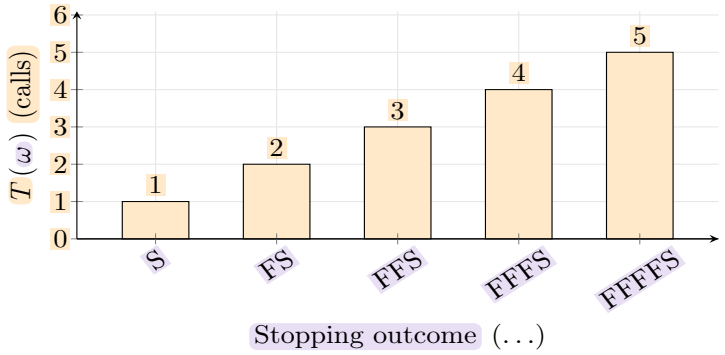
$$\sum_{t=1}^{m} 2^{-t} = (1 - 2^{-1}) + (2^{-1} - 2^{-2}) + \cdots + (2^{-(m-1)} - 2^{-m}) = 1 - 2^{-m}.$$

The partial sum approaches 1 as m grows. The probability of m consecutive failures is 2^{-m}, so the probability of never succeeding is zero.

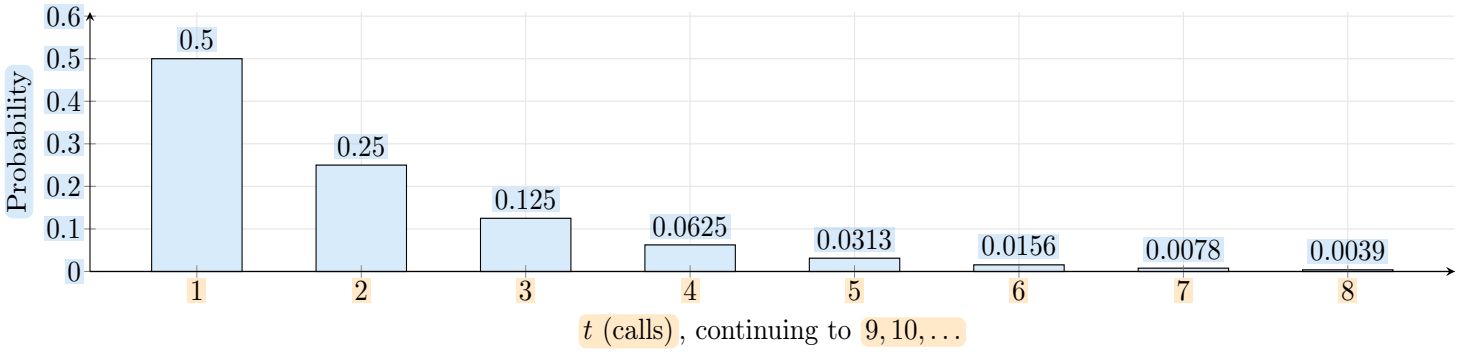
The graphs use Version 1. The first two panels preview the stopping outcomes of length one through five. Longer outcomes continue indefinitely.



Outcome probabilities (Version 1, finite preview). Later stopping outcomes have total probability 1/32. These bars sum to 31/32, without renormalization.



Assigned call counts (Version 1). Count every call, including the sale. Later outcomes map to $6, 7, \ldots$ calls; their omitted probability is $1/32$.



Waiting-count PMF (Version 1, first eight values). $p_T(t) = 2^{-t}$ for $t = 1, 2, \ldots$; the support continues indefinitely to the right. The omitted probability is $P(T > 8) = \sum_{t=9}^{\infty} 2^{-t} = 1/256$. Displayed bars total $255/256$ and are not renormalized; labels are rounded.

Task 3. Find and interpret a requested probability.

$P(T > 3) = 1 - \sum_{t=1}^{3} p_T(t) = 1 - (1/2 + 1/4 + 1/8) = 1/8 = 0.1250$. In each version, there is a 12.5% chance of needing more than three attempts to make the first sale, receive the first quote within budget, or draw the first qualifying invoice, respectively.

5.3 Calls until a sale

The preceding model found $P(T > 3)$. Here, use the same masses to find the complementary probability $P(T \leq 3)$.

A sales team calls until the first sale. Model each call as independent with sale probability $1/2$. Record each call as S for a sale or F for a failed call. The team models T, the call number of its first sale, by $p_T(t) = (1/2)^t$ for $t = 1, 2, \ldots$. It also records exact call duration D in minutes.

Questions/Tasks

C1 Task 1. Identify the outcomes and their probabilities. List the complete call sequences ending at the first sale and calculate their probabilities.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define T with units. Map each stopping sequence to its call count, list the support, and build and verify the waiting-count PMF. Contrast T with the exact call duration D.

C1 Task 3. Find and interpret a requested probability. Use the PMF to find $P(T \leq 3)$ and interpret it for the sales team.

Complete worked solution

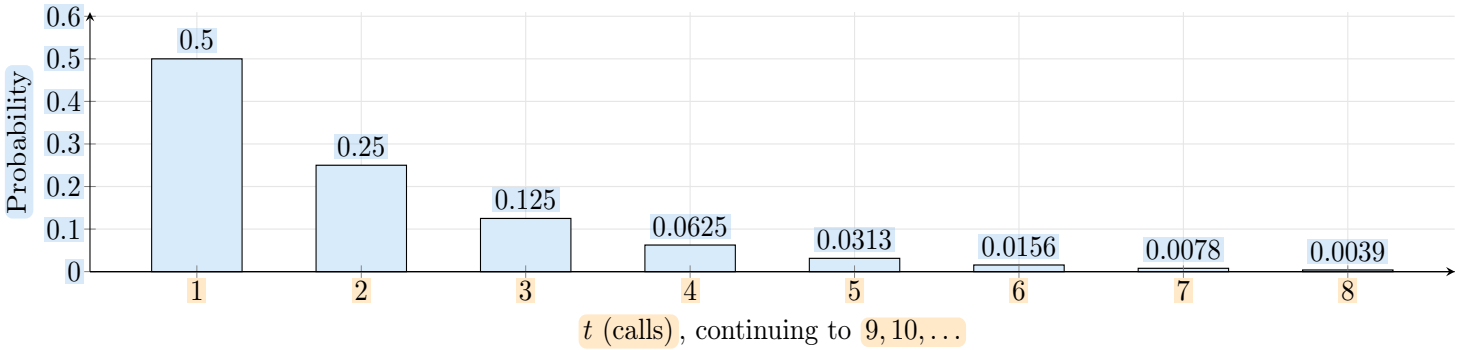
Task 1. Identify the outcomes and their probabilities.

The complete stopping outcomes are S, FS, FFS, ... The outcome ending on trial t is $t - 1$ failures followed by one success. Independence gives its probability as $(1/2)^{t-1}(1/2) = 2^{-t}$. Thus the first three outcomes have probabilities $1/2, 1/4, 1/8$.

Task 2. Define variables and units; map, list support, build and check PMFs.

T counts calls through the first sale. The sequence with $t-1$ failures and then a sale maps to t, so the support is $\{1, 2, \ldots\}$. Each count has one stopping sequence, giving the supplied $p_T(t) = 2^{-t}$. All masses are positive and the first m terms sum to $1 - 2^{-m}$, which approaches 1.

The count T is discrete. The exact call duration D, in minutes, is treated as continuous and can take values throughout an interval. The model gives no distribution for D.



Waiting-count PMF: first eight values only. $p_T(t) = 2^{-t}$ for $t = 1, 2, \ldots$; the support continues indefinitely to the right. The omitted probability is $P(T > 8) = \sum_{t=9}^{\infty} 2^{-t} = 1/256$. Displayed bars total $255/256$ and are not renormalized; labels are rounded.

Task 3. Find and interpret a requested probability.

Include the third call:

$$P(T \leq 3) = \sum_{t=1}^{3} p_T(t) = 1/2 + 1/4 + 1/8 = 7/8 = 0.8750.$$

The first sale occurs within three calls with probability 87.5%.

5.4 Delivery records: two numerical descriptions

Return to the idea from the school sale. One experiment can define several random variables.

A delivery has outcomes early, on time, late with probabilities $0.2, 0.5, 0.3$. A bonus B is 2 USD for early or on time and -3 USD for late. An indicator I is 1 if late and 0 otherwise.

Questions/Tasks

C1 Task 1. Identify the outcomes and their probabilities. Record the three delivery outcomes and the supplied probability of each.

C1 Task 2. Define variables and units; map, list support, build and check PMFs. Define B and I, with units where applicable. Map each outcome to both values, list the supports, and build and check both PMFs. Contrast the variables with an exact delivery-time measurement.

C1 Task 3. Find and interpret a requested probability. Use the bonus PMF to find $P(B < 0)$ and interpret it for the delivery.

Complete worked solution

Task 1. Identify the outcomes and their probabilities.

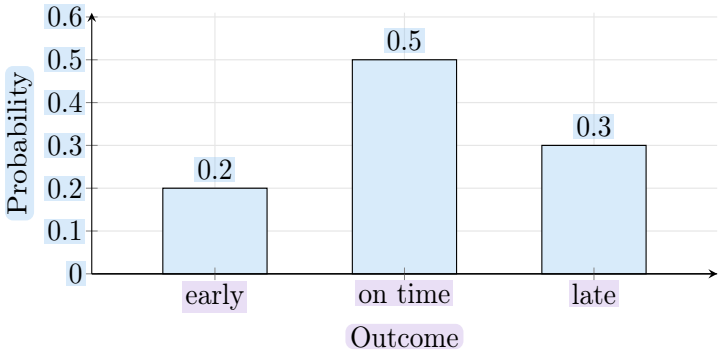
The complete outcomes are early, on time, and late, with probabilities 0.2, 0.5, 0.3 respectively. The three outcomes are mutually exclusive and their probabilities sum to 1.

Task 2. Define variables and units; map, list support, build and check PMFs.

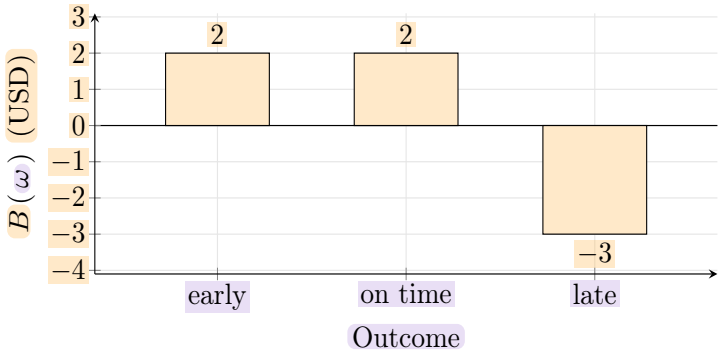
B is the bonus in USD with support $\{-3, 2\}$. The unitless lateness indicator I has support $\{0, 1\}$. Early and on-time deliveries both give $B = 2$ and $I = 0$. A late delivery gives $B = -3$ and $I = 1$. Grouping the first two outcomes gives

$$p_B(2) = 0.2 + 0.5 = 0.7, \quad p_B(-3) = 0.3,$$
$$p_I(0) = 0.2 + 0.5 = 0.7, \quad p_I(1) = 0.3.$$

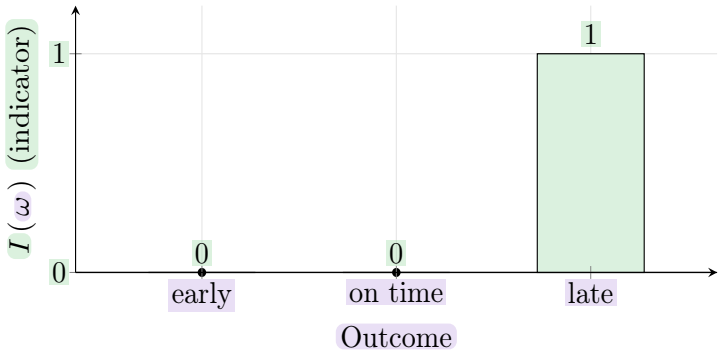
Each PMF has non-negative masses that sum to 1. Exact elapsed delivery time is continuous in an ideal measurement model. The early, on-time, and late categories alone give no exact durations or distribution of elapsed time.



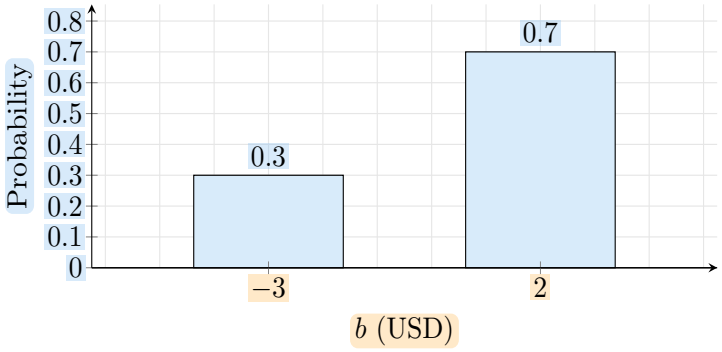
Delivery-outcome probabilities. The bars refer to delivery categories.



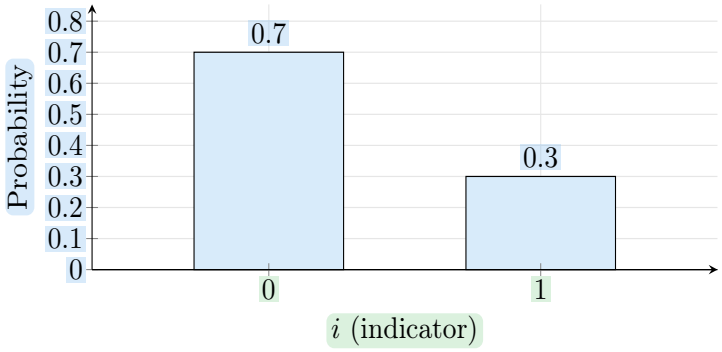
Assigned bonuses. Late delivery gives -3 USD, shown below zero.



Assigned lateness indicator. Only late delivery maps to 1; the indicator has no units.



Bonus PMF. Early and on time group at 2 USD; a negative bonus still has a positive probability.



Indicator PMF. Early and on time group at 0. Compare the outcome probabilities above.

Task 3. Find and interpret a requested probability.

The only negative bonus is -3 USD. The bonus PMF gives $P(B < 0) = p_B(-3) = 0.3000$. A delivery incurs a 3 USD deduction with probability 30%. This is the same event as $I = 1$.